Fig. 6 | Computational cost, read-depth stability and the VeloGIF framework. a, Runtime across 1,000–60,000 cells, with labels rounded up to minutes or hours and colors capped at 60 minutes. Blank cells indicate unavailable measurements. **b,** Mean per-cell velocity cosine relative to full-depth predictions (1.0× reads). The 1.0× point is a self-comparison; reduced-depth points assess prediction consistency, not biological accuracy. The cell2fate results use 9,144 shared cells and 472 shared genes. TopicVelo denotes integrated-transition displacement in fixed full-depth common-gene Ms coordinates, using a single-topic stochastic configuration. **c,** Summary of 19 method configurations across accuracy, scalability and stability. Stability uses the mean signed velocity cosine at 0.2, 0.4, 0.6 and 0.8× reads against 1.0× reads. The self-comparison at 1.0× is excluded. All 19 configurations have stability results. TopicVelo uses its integrated transition matrix projected into fixed full-depth common-gene Ms coordinates. This transition-projected measure differs from the direct gene-velocity cosine used for cell2fate and other methods. Overall Score = 0.6×Accuracy rank index + 0.4×Scalability rank index; stability does not enter this score. Accuracy is aggregated over available domains among eight, and coverage/preprocessing differ among methods. Circle area maps signed cosine/GDC to (x + 1)/2 for visualization; signed values and coverage are retained in Source_data. This frozen 19-configuration composite retains eight documented endpoint differences from later sources; the new PhyloVelo diagnostic is not included. **d,** Overview of the VeloGIF framework and included methods. Method coverage varies across datasets.